TUKZIE Vacation Work Programme 2026/2027

Autonomous Driving and Sensor Testbed Development
Duration: 6 Weeks (Two Compulsory Work Periods)
Project: Tukzie-Multidisciplinary E-Mobility Innovation Platform
Website: www.tukzie.com
Contact person: Sampath Jayalath



About the Project

Tukzie is a multidisciplinary engineering innovation platform focused on developing next-generation electric mobility solutions tailored to African environments.
We are inviting motivated engineering students to participate in a six-week vacation work programme focused on **autonomous driving technologies and the enhancement of Tukzie's existing sensing testbed**.
The programme will involve a single integrated engineering project with two interconnected technical focus areas. Students will work collaboratively to improve the vehicle's sensing capabilities and develop autonomous driving functionalities through hands-on design, implementation, integration, and testing.

Project Focus Areas

1. Autonomous Driving Development

- Design and develop electromechanical drive-by-wire systems to enable autonomous vehicle control.
- Develop mechanical steering mechanisms incorporating servo motors, actuators, and appropriate mechanical linkages.
- Investigate and implement electronic control of steering, acceleration, and braking systems.
- Design and integrate motor drivers, actuator control circuits, and embedded control systems.
- Develop closed-loop control algorithms for accurate steering, speed regulation, and vehicle manoeuvring.
- Develop and implement autonomous vehicle navigation, obstacle detection, collision avoidance, and path-planning algorithms.
- Integrate sensor feedback with actuator control to enable autonomous driving functionalities.
- Implement essential safety mechanisms, including emergency stops, actuator limits, manual override, and fail-safe operation.
- Validate mechanical, electrical, and software integration through simulations and controlled vehicle testing.

2. Existing Sensor Testbed Enhancement

- Evaluate the performance and limitations of Tukzie's existing sensing testbed.
- Improve sensor integration, calibration, accuracy, and reliability.
- Investigate sensor fusion techniques for improved environmental perception.
- Enhance real-time data acquisition, processing, visualisation, and logging.
- Develop a reliable sensing architecture that supports autonomous vehicle operation.
- Further improvement to digital twin architectures.

Both focus areas will be developed as part of a **single integrated autonomous mobility system**, allowing students to contribute across hardware, software, sensing, control, and systems integration.

Expected Outcomes

By the end of the programme, participating students are expected to deliver:

- An improved and documented sensing testbed.
- Implemented and demonstrated autonomous driving functionalities.
- Integrated and tested hardware and software components.
- Evaluation of system performance, accuracy, reliability, and limitations.
- Technical documentation, source code, and a final project demonstration.

Who Should Apply?

We welcome enthusiastic students from:

- Electrical Engineering
- Mechanical and Mechatronic Engineering
- Electrical and Computer Engineering
- Computer Science or AI
- Other relevant engineering disciplines

Experience in Python, C/C++, ROS/ROS 2, embedded systems, robotics, computer vision, or sensor integration will be advantageous.
Students should demonstrate strong problem-solving skills, an interest in intelligent mobility, and the ability to collaborate within a multidisciplinary engineering team.

Compulsory Commitment: Non-Negotiable

The vacation work programme requires a full six-week commitment across BOTH periods:
Period 1: 23 November-10 December 2026 (3 weeks)
Period 2: 18 January-5 February 2027 (3 weeks)

Participation in BOTH periods is COMPULSORY and NON-NEGOTIABLE. Vacation work completion letters will be issued to candidates who successfully complete the full six-week programme.

Students must be available for the entire duration of both periods. Partial participation, late commencement, early departure, or attendance during only one period will not be accepted.
Applicants must confirm their full availability for both periods before applying. If interested, please email Dr Sampath Jayalath (sampath.jayalath@uct.ac.za).